Engineering Intelligence Product Development Roadmap

Prepared for Muhammad Assim Baig | 18 October 2026 | Planning baseline 1.0

Deliver three usable workflows on one governed platform

Telecom survey, telecom inspection and construction inspection share Site identity, evidence, controlled templates, independent decisions and reports. Preserve the existing engineering lifecycle and expand it in stages. Complete a bounded field pilot before claiming full operational readiness.

Immediate product	Full lifecycle target	Delivery principle
Telecom surveys and inspections; construction inspections; guided role homes; reliable photo evidence and review.	Survey to design, BOQ, procurement, construction, acceptance, asset handover, maintenance, invoices and closeout.	Reuse current implementation. Distinguish source present, automated checks, deployed behavior and real-user acceptance.

Verified source position	What is still unverified
main at 114ccf3e1273; PR129 merged 9 October 2026. Latest inspected Quality checks workflow succeeded for that commit.	This planning session did not run the app, inspect live deployment receipts, perform browser/mobile QA or validate actual engineering outputs.
Repository manual 3.39 records live baseline 44 and local work through original Stage 5. Saved manual 3.22 is older.	Manual 3.39 still says PR129 awaits merge; GitHub merge evidence supersedes that statement only. Publication and human acceptance remain separate.

Delivery files: this PDF contains the requirements and workflow atlas; the companion .drawio contains 11 editable diagram pages. These are Visio-style diagrams, not a native .vsdx file. No application code or production setting is changed by this roadmap.

Evidence and current development gaps

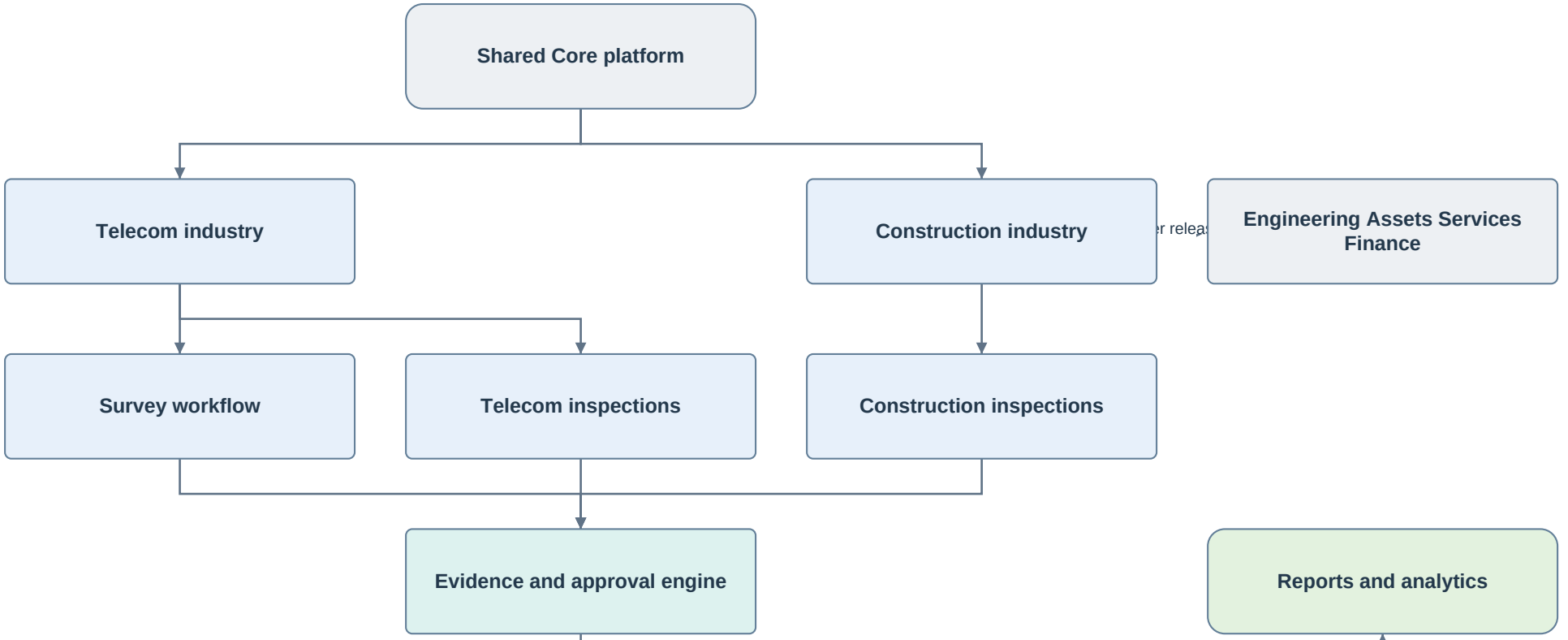
Treat this as a planning audit of selected current records, not a fresh end-to-end software acceptance audit.

Original stage	Evidence on current main	Required next action
0 Baseline and recovery	Documented local backup, startup and recovery work.	Reproduce the environment and current rollback package; verify the user PC state separately.
1 Role homes and Site workspace	Source merged; local role-response evidence retained.	Desktop and mobile usability review; remove untraceable navigation and overloaded pages.
2 Survey	Capture, saved revisions, readiness and independent review documented.	Actual field-device upload, three-person review and report acceptance; AI remains unqualified.
3 Engineering and release	Implemented/API-verified per records; bounded calculations.	Browser and qualified engineering/customer checks; do not claim full tower design or CAD generation.
4 Commercial	BOQ, sourcing, PO and variations implemented/API-verified locally.	Procurement authority, real quantity reconciliation and browser acceptance.
5 Construction	Local implementation includes ITP, inspections, NCRs, commissioning and dossier.	Actual construction pilot; check serials, measured quantities, hold points and customer acceptance.
6 Assets / 7 Services / 8 Finance	Queued in original stage register; not operationally completed.	Implement separate domain authority, records and receiving/financial acceptance.
9 Acceptance and rollout	Queued; earlier gates remain open.	Bounded pilot acceptance first, then full lifecycle acceptance when Stages 6-8 exist.

Repository status summary: 7 packages and 56 logical subfeatures; inventory does not prove completion. Historical test counts are evidence reported by the project, not tests rerun in this session.

Product structure and industry scope

Industry is a project/site context. Workflow type chooses the template and required evidence; it must not create a second disconnected evidence store.



Blue = work; teal = review/control; amber = decision; green = output; grey = shared input or later scope. Construction survey is outside the requested first release. Existing engineering features remain available only within their actual supported scope.

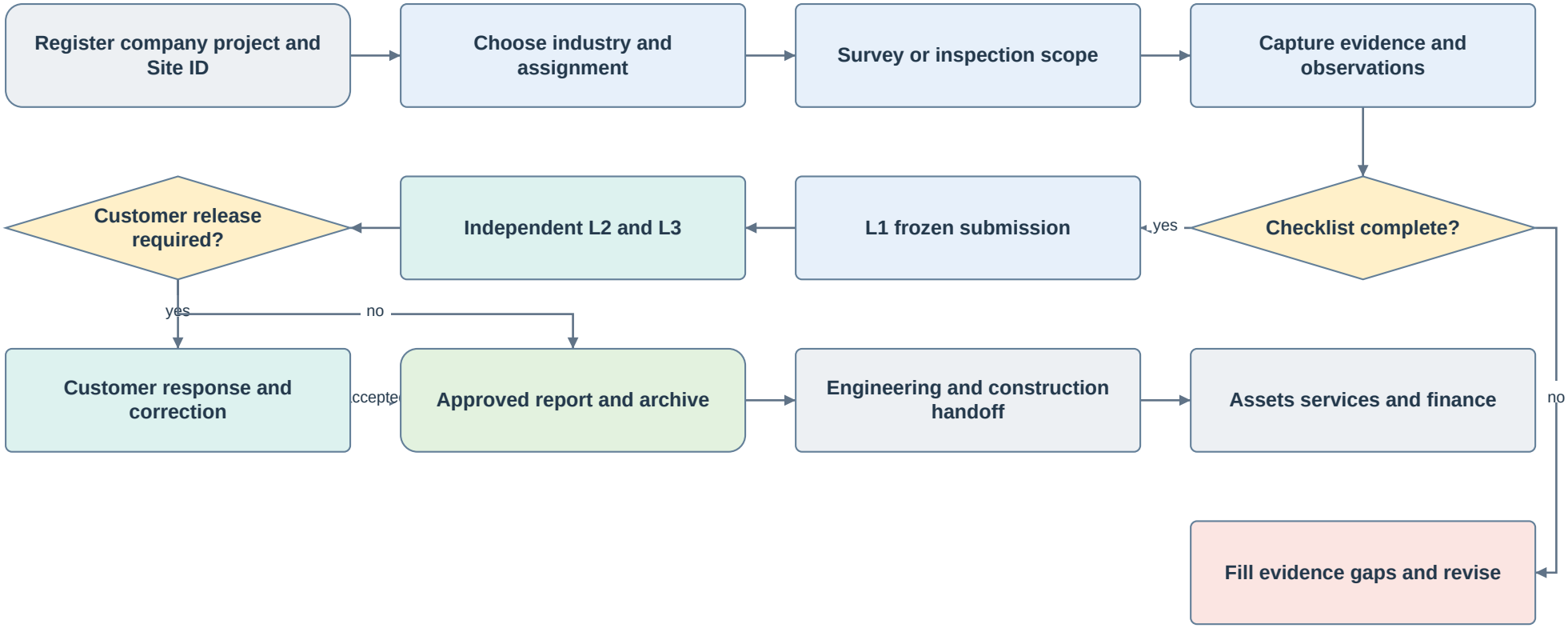
Navigation and user interface requirements

Stage 1 / 1.1-1.9 remains the navigation foundation. The new industry selector must be added compatibly rather than replacing established stage numbering or permissions.

Area	Required behavior	Acceptance condition
Main navigation	My work; Projects and sites; Review queue; Reports; Administration when permitted. Show an industry filter with Telecom / Construction.	A field engineer can find an assigned Site and start the correct workflow without opening unrelated modules.
Persistent Site workspace	Visible Site ID, project, industry, lifecycle position, current revision, responsible people and next action. Preserve context between tools.	Switching tools retains Site context. Switching Sites with unsaved work prompts a clear save/discard choice.
Working layout	List/map switch; searchable left navigation; central record; contextual evidence/review panel. Collapse panels on phones.	A narrow screen remains usable without horizontal scrolling for primary tasks; no hover-only actions.
Workflow guidance	One clear next action, missing prerequisites and blocking reasons. Separate task progress from physical construction progress.	No fabricated completion percentage; counts reconcile with accessible records and filters.
Record states	Loading, empty, draft, saved, returned, approved, stale, unavailable, forbidden and partial upload must be distinct.	Error messages identify the action and recovery path; network failure is never shown as zero work.
Administration	Normal Admin maintains organisation lists; authorised PM maintains project options. Stable codes and historical meanings.	No Super Admin dependency for routine catalogs; inactive values remain visible in history.

End to end product workflow

One source of truth links every field observation, finding, document and approval to the registered Site ID.



Customer release is conditional on contract and workflow. Internal approval does not mean customer acceptance. Later engineering, asset, maintenance and finance consumers must reference the exact approved source revision.

Roles and authority matrix

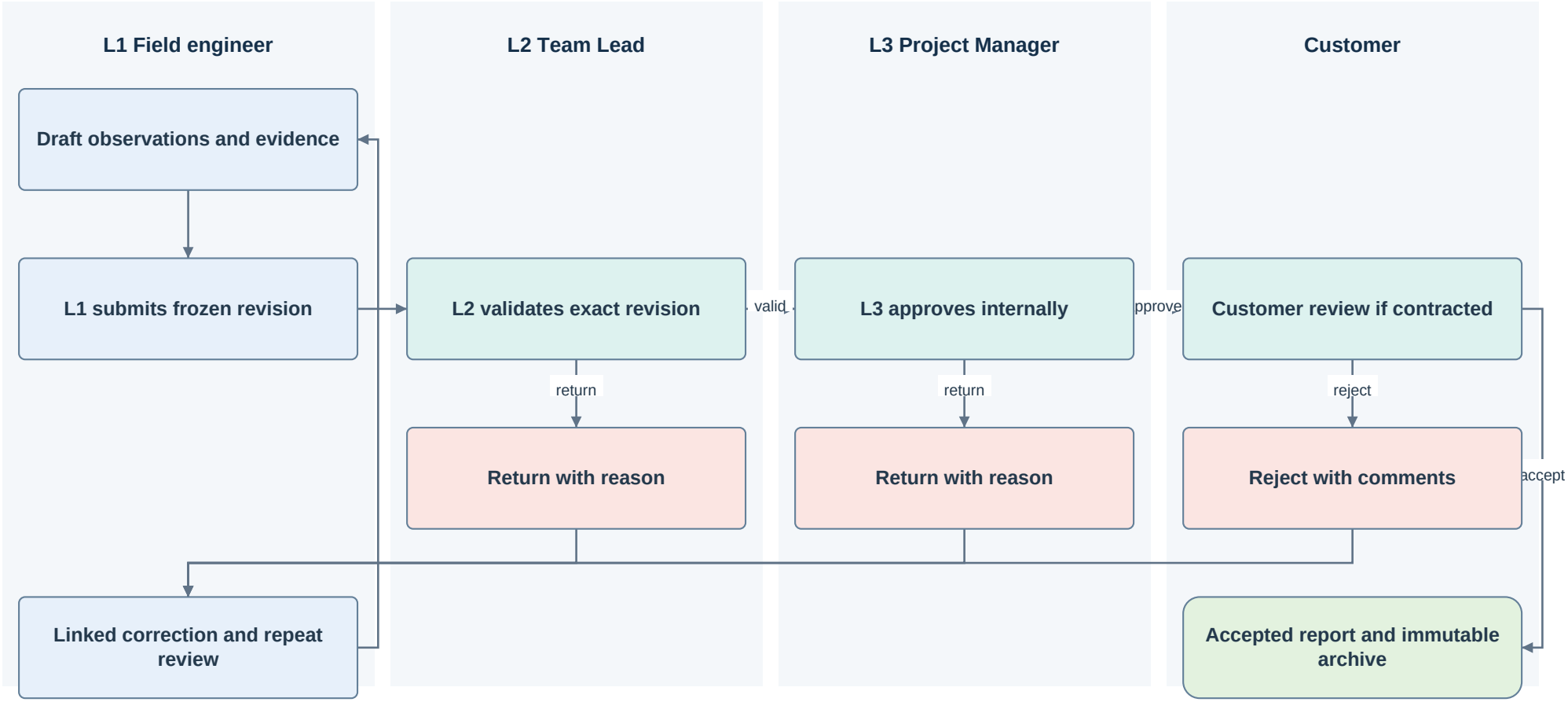
Stage 1 / role homes and Stage 9 / 9.1 must enforce this model on APIs, exports and original-file access, not just visible buttons.

Role or layer	Permitted responsibility	Limits to preserve
Super Administrator	Platform configuration, organisation administration and permitted audit overview.	No engineering or financial approval from administrative privilege.
Organisation Admin	Manage allowed non-admin accounts, assignments and organisation catalogs.	Own organisation only; cannot self-escalate or infer technical authority.
Project Manager	L1 preparation, L2 validation, L3 approval and L4 overview only where explicitly assigned.	Different authenticated people must submit, validate and approve the same revision.
Team Lead	Assigned preparation, independent L2 review and permitted overview.	No L3 approval; team membership alone does not grant Site access.
Engineer	Assigned capture, draft, submission and correction; permitted overview.	No independent validation or approval of governed records.
Contractor / customer / oversight	Contractor submits; assigned customer reviews/releases; oversight reads, all within named-role ceilings.	Customer release authority is separate; oversight cannot mutate. New domain grants require explicit reviewed mapping.

Required access dimensions: organisation, department, project, team, Site, domain and workflow level. Role is a ceiling on grants. Revocation, expiry, cross-tenant denial and direct-file URL access are mandatory negative tests.

Submission validation approval and returns

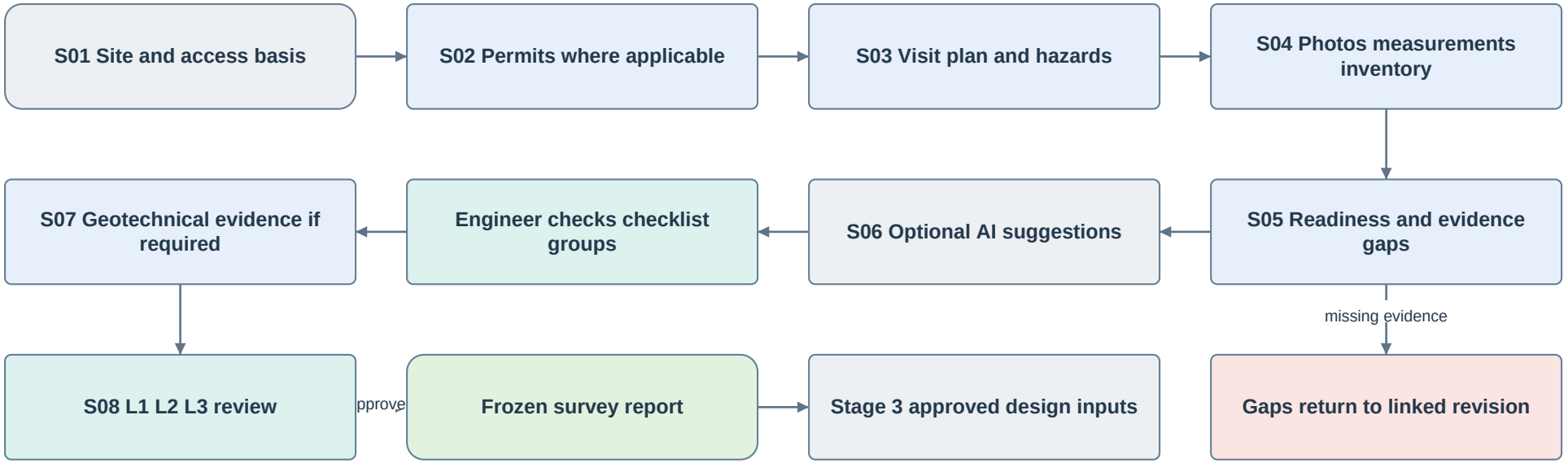
Common governance for Survey, Engineering and Construction. L4 is read-only overview, not an additional approval stage.



Any returned or rejected submission creates a reason and linked correction. Old snapshots stay immutable. Customer-optional branches require an explicit applicability record; this diagram shows the contracted customer-review path.

Telecom survey workflow

Stage 2 / 2.1-2.5 uses existing S01-S08 capabilities and closes actual field acceptance gaps.



Acquisition, permits and geotechnical inputs are required only where the responsible reviewer determines applicability. Unknown measurements remain unknown. Optional AI failure must leave the manual workflow usable.

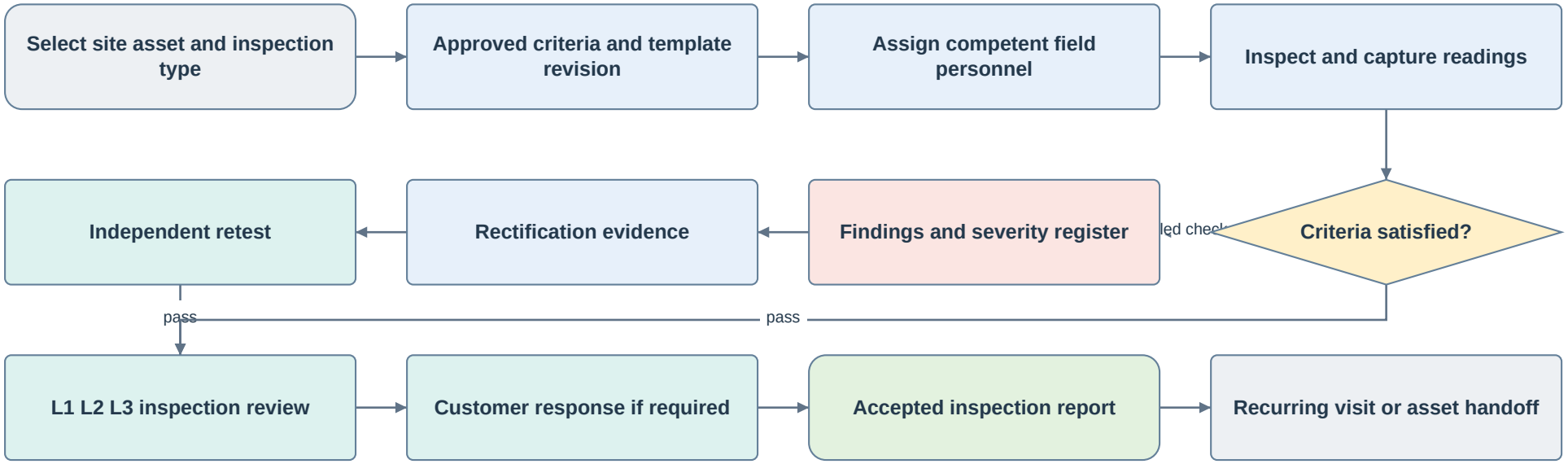
Telecom survey functional requirements

Initial release scope: register the Site, collect traceable evidence, complete a controlled survey and issue a reviewed report.

Requirement ID	Required behavior	Definition of done
SUR01 Site context	Approved project, registered Site ID, coordinate datum/source and assigned engineer. Prevent accidental upload to another Site.	Denied foreign Site access; explicit Site confirmation; no guessed coordinates.
SUR02 Template	Versioned sections and questions with evidence needs, units, applicability, mandatory status and approved criteria.	Existing 150 requirements across 11 sections preserved where applicable; new template is a controlled revision.
SUR03 Capture	Site access, tower type, dimensions, visible condition, power/DC equipment, earthing observations and inventory as applicable.	Readings have units, source/measurement method and evidence; no values inferred merely from photos.
SUR04 Evidence	Multiple photos per item and one photo linked to multiple applicable items, with annotation on a derived preview.	Original unchanged; saved link identifies photo, checklist item, asset/location and author.
SUR05 Readiness	Show missing, blurred/unreadable, unavailable and unreviewed evidence separately.	Readiness checks do not claim engineering adequacy or prove asset absence.
SUR06 Revision and report	Explicit saved drafts; immutable submitted versions; independent review; report and machine-readable export.	The final report matches the exact saved/approved revision and includes gaps, exclusions and decisions.

Telecom inspection workflow

Use a controlled inspection type: installation acceptance, tower condition, civil quality, equipment or maintenance inspection. Activate only templates that have been technically reviewed.



A failed inspection may be recorded and reviewed; it must not release blocked work. Missing evidence does not equal pass. Photographs cannot establish hidden conditions, torque, concrete strength or earthing resistance without the required measurement evidence.

Telecom inspection functional requirements

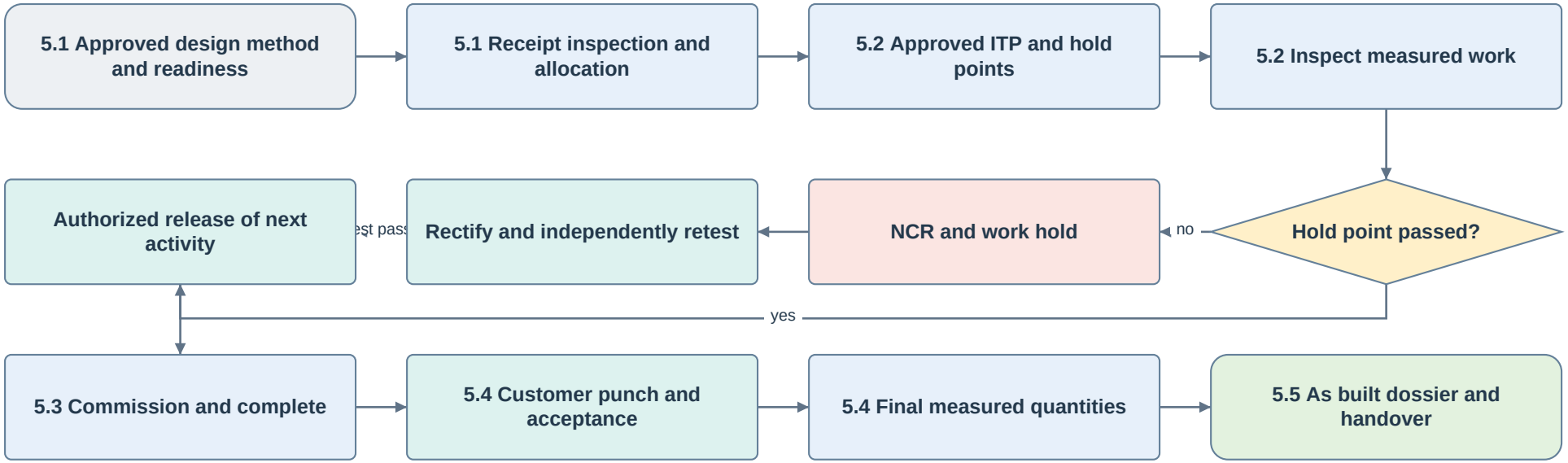
Industry extension TI is a roadmap overlay mapped to Stage 2 capture, Stage 5 inspection/NCR and later Stage 7 maintenance. It does not renumber original stages.

Requirement ID	Required behavior	Acceptance condition
TI01 Scope and asset	Inspection type, asset/component identity, visit purpose, location and approved scope.	A tower, compound or equipment inspection has an unambiguous Site/asset reference.
TI02 Criteria registry	Record client specification and applicable standard/edition; clause references, acceptance limits and approved method.	Qualified reviewer confirms applicability; software never substitutes a generic checklist for project criteria.
TI03 Measurements	Instrument identity, calibration validity, observed value/unit, time, operator and source evidence where measurement is required.	A passing numeric check contradicting its recorded limit is rejected. Expired calibration blocks relevant acceptance.
TI04 Defects and findings	Type, severity, affected asset, observation, evidence, owner, due date and blocking status.	Open critical/blocking findings prevent acceptance; administrative closure cannot clear engineering defects.
TI05 Rectification and retest	Original finding remains; remedial evidence and independent retest are linked as revisions/events.	Closure requires the defined competent authority and acceptable retest evidence.
TI06 Reports and recurrence	Client-format report; approved reference evidence; status history; repeat visit linked to earlier findings.	Report can be reconstructed later; recurring inspection does not silently reuse earlier pass results.

TIA, ACI and project specifications belong in a controlled reference registry. Standards are not interchangeable; full copyrighted text or AI ingestion needs appropriate rights. The app does not issue professional or statutory approvals by itself.

Construction inspection workflow

Stage 5 / 5.1-5.5 already has a merged implementation foundation. The next work is real-browser usability, project qualification and controlled adoption.



Hold/witness requirements come from an approved ITP. Readiness for receiving and readiness for physical work are separate. An approved failed-inspection record is still failed; authorised hold-point release requires the defined passing or reviewed exception evidence.

Construction inspection requirements

Use the same governed engine for residential, commercial and telecom civil works, while retaining the project-specific ITP and specification.

Stage and requirement	Required detail	Acceptance condition
5.1 / CON01 Readiness	Approved design, access, method, personnel competence, work window and material basis. Existing stock is a separate authorised source.	Receiving authorisation cannot release construction; required documents remain current.
5.1 / CON02 Materials	Delivery, source PO or approved stock, lot/serials, accepted/rejected/quarantined quantities and storage.	Received = accepted + rejected + quarantined. Allocation cannot exceed eligible stock.
5.2 / CON03 ITP and inspection	Inspection request, activity/location, approved ITP, criteria, hold/witness points, readings, calibration and photos.	Missing or failed required tests block the affected release. Approved non-applicability requires recorded authority.
5.2 / CON04 NCR	Finding, severity, root cause where required, action owner, remedy, retest and closeout decision.	No closing a finding solely by uploading a photo; criteria and independent verification apply.
5.3 / CON05 Completion	Installed serials, commissioning/test evidence, measured completed work and signed declaration.	No foreign or unallocated serials; required failed/missing tests block completion.
5.4-5.5 / CON06 Acceptance	Customer punch list; joint final measurements; as-builts, certificates and dossier applicability.	Certified quantity cannot exceed measured eligible completed work or approved scope; blocking defects remain visible.

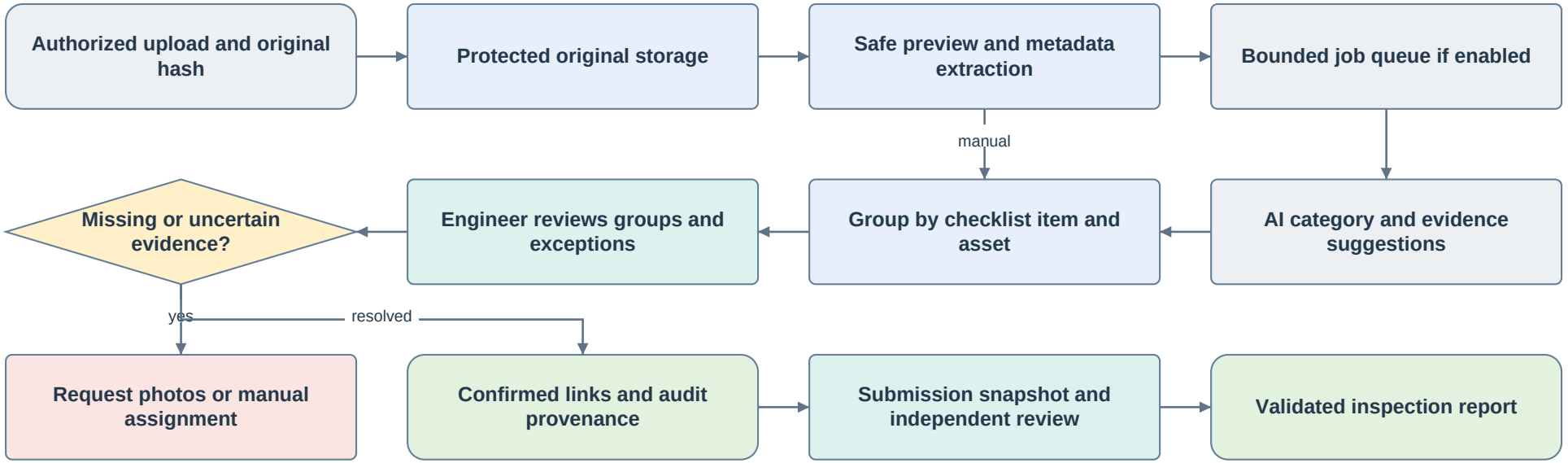
Construction template library and configuration

Templates are reusable controlled records. Publishing a template does not make it appropriate for every project.

Template group	Examples of required evidence	Configuration rules
Earthworks and foundations	Excavation level, founding stratum acceptance, replacement material, lift thickness, compaction test and waterproofing where specified.	Use project-defined limits and geotechnical recommendations; no generic bearing-capacity inference.
Reinforcement and formwork	Bar size/grade, spacing, cover, laps/couplers, supports, formwork dimensions and embeds before pour.	Reference approved drawings and revision; retain concealed-work hold points.
Concrete works	Mix/submittal, delivery ticket, site tests, sampling, placement, curing and laboratory results.	Concrete strength acceptance requires the specified tests and qualified interpretation, not photo classification.
Masonry and finishes	Material approval, setting out, dimensions, alignment, joints, waterproofing, finishes and snagging.	Trade and activity-specific criteria; distinguish observation from measured acceptance.
MEP and safety	Concealed services, installation records, continuity/pressure/functional tests, calibrated instruments and safety checks.	Activate only with competent discipline ownership; safety stop-work is an immediate governed escalation.
Template publishing	Draft template, technical review, approved revision, project selection and deactivation.	Edits do not rewrite completed inspections; each record freezes the template and criteria revision.

Photo analysis and checklist matching

S06 remains advisory and unqualified in the inspected baseline. This is the target pipeline, with a manual path required at every AI-dependent step.



Engineer review is organised by checklist item, asset and exceptions. No bulk action may convert uncertain labels into an engineering pass. AI recommendations retain model/version, input digest, time, confidence and human disposition.

AI requirements and practical review workload

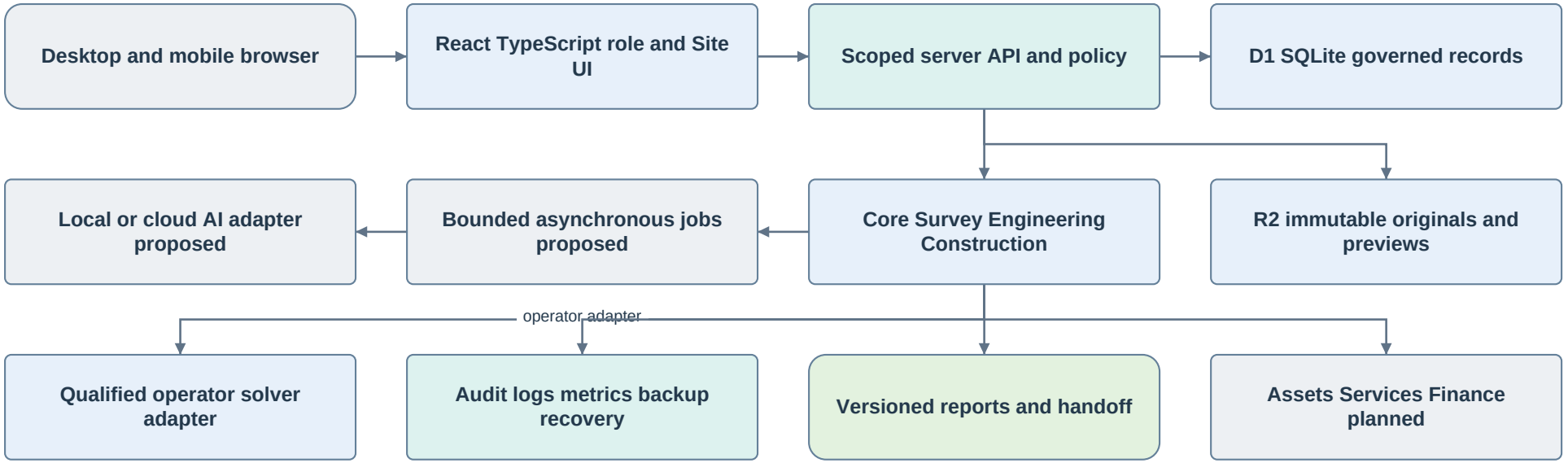
The product must reduce evidence sorting without promising that every technical conclusion can be automated.

Requirement	Implementation detail	Release gate
AI01 Evidence grouping	Suggest categories and candidate checklist links; show representative images, coverage and unassigned evidence.	Evaluate on a labelled dataset from each released workflow and actual field image conditions.
AI02 Exceptions first	Prioritise uncertain labels, conflicting evidence, low-quality images, duplicates and missing mandatory views.	Reviewer can filter to exceptions, open all originals and undo any accepted link.
AI03 Human confirmation	Engineer confirms/rejects suggested groups and links; L2/L3 independently decide governed submissions.	Human confirmation records actor/time/revision. A suggestion never becomes approval automatically.
AI04 Bounded execution	Job state, per-file retry, timeout, size limits, idempotency, cancellation, cost cap and model/provider provenance.	Provider unavailable or quota exceeded leaves manual work available; no repeated charge for identical completed task without reason.
AI05 Quality metrics	Measure per-class precision/recall, incorrect links, unresolved items and engineer minutes per report.	Agree thresholds before trial; report false negatives and critical misses separately. No invented accuracy claim.
AI06 Provider strategy	Start with one narrow classification task. Compare local and cloud options using real photos, privacy and total operating cost.	Provider approval and data-use terms precede external transfer. AI deployment is a separate accepted feature.

200 photos is the current documented selected-survey limit, not a throughput qualification. A later 1,000-photo target needs storage, queue, pagination, review and field-network tests before it is promised. Missing/low-confidence evidence must not be hidden by a batch action.

Architecture and integration boundaries

Keep the existing stack and modular ownership. A new queue or external provider requires a demonstrated requirement and reviewed implementation.



React/TypeScript, Vite/vinext, Worker, D1/SQLite and R2 reflect inspected repository guidance. Arrows show logical contracts, not automatic execution or certification. Operator solvers remain separately qualified; local AI and cloud AI are alternatives, not both mandatory.

Data model and master data requirements

These are logical contracts, not an instruction to create all tables immediately. Reuse and extend existing versioned records after schema review.

Contract	Required facts	Integrity requirement
Organisation and access	Identity, layer, role, assignment scope, expiry, issuer and explicit domain/level.	Deny by default; revoke promptly; no cross-tenant aggregation or file leakage.
Project and Site	Stable IDs, industry, customer, region, source coordinates/CRS, ownership and lifecycle references.	Names may change through review; identities and history remain traceable.
Template and criterion	Template/version, section/question, response type, units, evidence needs, applicability, criteria source and reviewer.	Freeze the exact revision per inspection; deactivate rather than destroy referenced options.
Evidence and links	Original ID/hash, bytes/type/size, uploader/time, Site, source EXIF and derived-preview provenance.	Original immutable; explicit many-to-many item links; signed or authenticated retrieval.
Inspection and finding	Workflow type, asset/location, observed data, method/instrument, findings, actions, due dates and retests.	No silent default pass; distinguish not applicable, not inspected, failed and passed.
Revision and decision	Snapshot ID/digest, prerequisites, state, independent actors, reasons and ordered events.	Changed inputs invalidate affected progression; never rewrite old approvals.
Handoff and reporting	Source revision, scope, limitations, evidence manifest, customer state and receiving acceptance.	Receiving work does not inherit approval automatically; export enforces current access policy.

Admin/PM catalogs should include configured customer, region, asset/tower type, equipment, document template/revision, standards/editions, materials, units, inspection types and defect codes where approved. Avoid free-text duplicates in repeated business fields; retain narrative for real observations and assumptions.

Reliability security and capacity requirements

Proposed acceptance targets below are planning targets to agree and measure. They are not claims about current performance.

Area	Requirement and proposed target	Proof required
Uploads and originals	At least the existing 200-photo selected workflow; per-file progress and retry; no lost successful files; explicit size/type limits.	Actual supported phone/network run, interrupted upload and mixed-success retry; compare original hashes.
Performance	Pilot target: common list/search p95 under 2 s; normal record load under 3 s; dashboard under 5 s at an agreed pilot load.	Define dataset and concurrent users first; measure server and browser timings separately. Heavy AI uses jobs.
Offline and drafts	First release offers honest save status and safe interrupted-upload recovery. Full offline capture/sync is a later explicit feature.	If added: device privacy, encryption, conflict resolution, sign-out cleanup and multi-device tests.
Security	Server scope, account/session controls, secure secrets, file validation, bounded parsing, tenant separation and audit events.	Negative role/scope tests, original URL checks, dependency review and independent security review before enterprise rollout.
Recovery and operations	Separate database and original-file backups; tested restoration; monitoring, error alerts and rollback instructions.	Proposed pilot RPO 24 h / RTO 8 h only after tested and accepted; tighten for enterprise contracts.
Accessibility	Keyboard operation, visible focus, readable contrast, labelled inputs, touch controls and useful error summaries.	Manual desktop/mobile checks at agreed viewports plus actual field devices. HTTP 200 is not usability evidence.

Stage 3 engineering and Stage 4 commercial roadmap

Preserve original substages and current scope limitations. Source is already present for much of this work; acceptance and targeted repair come before broad new engines.

Original substage	Requirement and next work	Acceptance gate
3.1 Design basis	Current approved survey, applicable investigation/access, criteria editions, units, source originals and independent actors.	Stale or changed prerequisites block release; incomplete drafts remain possible.
3.2 Discipline evidence	Drawings/reports and bounded calculations; freeze input model and link results without circular dependencies.	Qualified reference comparisons; tool/version/environment and applicability retained; no full tower capacity claim.
3.3 Coordination and FDED	Discipline conflict register, blocking issues and frozen deliverable package.	Unresolved blocking coordination prevents release; source digests match.
3.4-3.5 Customer design release	Transmittal, customer response, linked correction, resubmission and archive.	Different actors and customer authority verified on real browser; archive preserves accepted history.
4.1-4.2 Quantities and baseline	Physical measured scope, units, takeoff, dated rates/currency and frozen costing.	No duplicate physical quantity; reconcile against a real checked BOQ and source dimensions.
4.3-4.5 Sourcing PO variations	Separate technical/commercial evaluation, authorised award and PO issue; controlled scope changes.	Procurement authority and commitment limits; concurrency/expiry tests; technical approval does not authorise spending.

Stage 5 implementation acceptance backlog

Stage 5 is recorded complete locally through issue124 and merged in PR129. Reuse its implementation and verify these high-value cases before production reliance.

Original substage	Tasks preserving T1 T2 T3 structure	Required evidence
5.1 Readiness and materials	T1 verify work/receiving basis; T2 receive and allocate purchased or approved existing stock; T3 test holds, stock limits and scope denial.	Real source PO/stock authority; reconciled lot quantities; no foreign serials.
5.2 Progress ITP and QA	T1 approve criteria/calibration; T2 inspect, log NCR, rectify and retest; T3 test failed/missing/held work.	A passing inspection and a failed inspection; failed record retained but release blocked.
5.3 Commissioning completion	T1 define acceptance tests; T2 measure actual installation and complete; T3 reject unsupported serial/test claims.	Installed serials derive from actual allocations; measured work and required results agree.
5.4 Customer and final measure	T1 inspect customer criteria/punch/BOQ; T2 rectify and jointly measure; T3 test stale/overclaimed quantities.	Claimed, measured and certified remain separate; eligible actual completion bounds certification.
5.5 Dossier and handoff	T1 preserve as-builts/certificates/manuals; T2 independently review dossier; T3 run complete browser pilot and restore.	Receiver-ready dossier; explicit applicability; original hash checks; preserved correction history.

Known project records include prior serial, quantity and duplicate-scope fixes. Their historical passing automated checks do not replace actual field acceptance. Any remaining original audit findings stay in the backlog until evidenced closed.

Stage 6 assets and handover

Original issue125 / 6.1-6.5 remains queued. First deliver a reliable asset register and receiving acceptance; advanced digital twins follow qualified source data.

Original substage	Required functionality	Acceptance gate
6.1 Asset identity	Create accepted Site-linked asset identity, tag, serial, model and source acceptance reference.	No duplicate active identity/serial or cross-Site attachment; QR/barcode may be a later UI addition.
6.2 Hierarchy and custody	Parent/child relations, location, custodian, movements and condition evidence.	No hierarchy cycles or simultaneous incompatible custody; observed condition is distinguished from inference.
6.3 Models and GIS	Model originals, revision, CRS/datum, processing provenance and accuracy limitations.	No claim of surveyed accuracy from photo GPS; unsupported files bounded/rejected; existing prototype not promoted silently.
6.4 Receiving acceptance	Dossier, receiver identity, spares, training, open obligations and explicit response.	Rejection and resubmission preserved; internal construction approval cannot imply receiver consent.
6.5 Modification and retirement	Link design/condition change, custody, disposal authority and retained obligations.	History and serial trace retained; retirement does not erase warranty, liability or evidence.

Stage 7 managed services

Original issue126 / 7.1-7.5 remains queued. Define service authority and accepted assets before implementing recurring work.

Original substage	Required functionality	Acceptance gate
7.1 Contract and onboarding	Signed service scope, accepted asset/site basis, SLA calendar, coverage, exclusions and competent team.	No invented service dates or entitlement; reviewed access and domain grants.
7.2 Plan and dispatch	Preventive schedules, faults, visits, workload, reassignment and permitted dispatch.	Dispatch creates an assignment, not evidence that maintenance occurred.
7.3 PMR and CMR	Readings, inspection evidence, fault diagnosis, repair, retest and restoration decisions.	Preventive visit approval cannot close an unresolved corrective fault. Preserve actual chronology.
7.4 Spares and custody	Issue, return, consumed parts, serial replacement and quarantine linked to work order.	No overissue; removed/installed serials and custody reconcile. No implicit ERP or IoT integration.
7.5 SLA and service evidence	Use measured incident intervals, agreed calendar and exclusions to prepare service/billing inputs.	Missing data remains explicit; no uptime percentage inferred from the absence of reports.

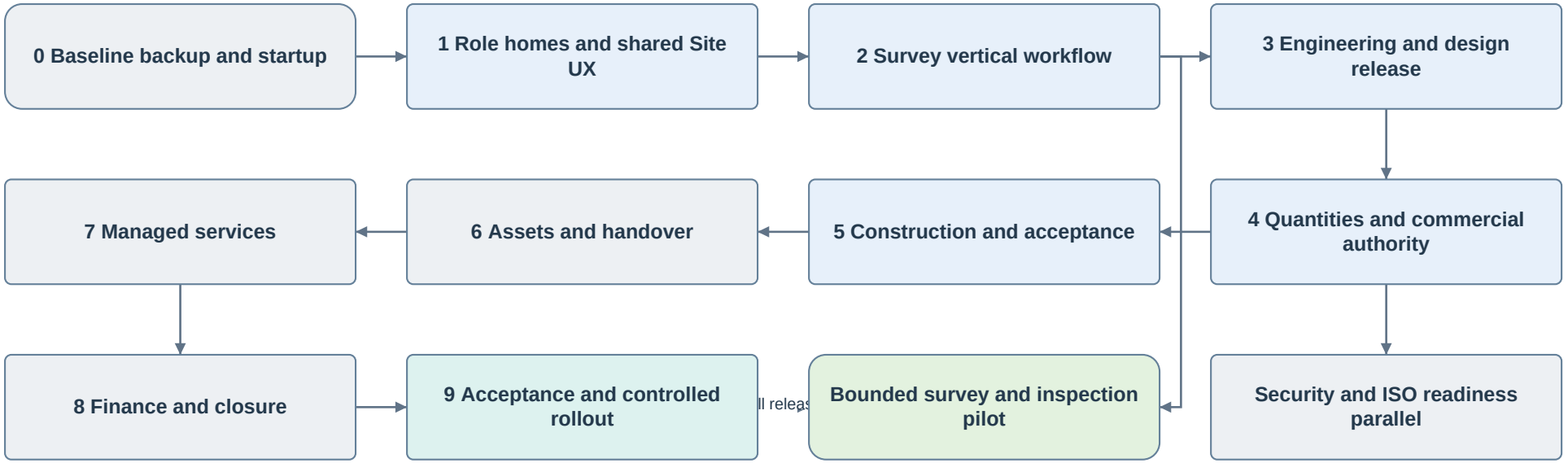
Stage 8 finance and closure

Original issue127 / 8.1-8.5 remains queued. Build evidence-backed records first; bank execution and tax-system integration are separate projects.

Original substage	Required functionality	Acceptance gate
8.1 Invoice and certification	Contract, approved eligible works/service sources, invoice identity, currency and declared tax treatment.	Reject duplicate invoices, unsupported quantities, inconsistent currency and missing sources.
8.2 Financial authority	Independent payable validation/approval, amount limits, delegated authority and expiry.	Technical approval and Admin access never confer finance authority.
8.3 Payment and reconciliation	Remittance evidence, partial payments, allocations, balances and controlled reversal.	Do not label paid until evidence is recorded; concurrent allocations cannot exceed payable balance.
8.4 Retention and warranty	Obligation dates, guarantees, defects, release criteria and reviewed non-applicability.	Unmet conditions block release; guarantee records need not wait for payment.
8.5 Final account and closure	Reconciled balances, claims, liabilities, dossier and customer/finance closeout.	Open liabilities and blocking defects remain visible; archived financial history stays immutable.

Development dependencies and release scope

Original Stage 0-9 numbering is retained. Industry-specific inspection requirements are overlays on existing capture, QA, reports and acceptance.



A bounded Stage 9 survey/inspection pilot can run before all future packages exist. It must disable unreleased packages and state its limits. Full lifecycle acceptance waits for the applicable Stages 6-8 and complete end-to-end evidence.

Proposed seven day pilot readiness sprint

Dated planning window: Sunday 18 to Saturday 24 October 2026. This is a bounded readiness sprint, not a promise to finish the full lifecycle in one week.

Date and original stage	Work package	Reviewable output and owner
18 Oct / 0 and 9.1	Confirm main/local state, recoverable backup and pilot scope; reconcile stale PR129 documentation.	Developer: baseline receipt. Assim: select telecom survey, telecom inspection and construction inspection samples.
19 Oct / 1.1-1.9 and 9.3	Walk five role homes and Site navigation; agree industry/task selector; fix pilot-blocking navigation only.	Developer + QA: desktop/mobile issue list and agreed screen flow.
20 Oct / 2.1-2.5	Test photo capture, partial retry, checklist mapping, saved revision and report on a real device.	Field engineer + QA: one complete manual telecom survey with original hashes.
21 Oct / 5.2 and TI overlay	Configure reviewed telecom inspection template; run pass, failed finding, remedy and retest.	Technical reviewer: criterion/evidence trace. Developer: gap fixes where reuse is insufficient.
22 Oct / 5.1-5.5	Run one bounded construction inspection and hold/NCR case; verify dossier/report path where applicable.	Construction reviewer: accepted pilot scenario and explicit exclusions.
23 Oct / 9.1-9.4	Independent accounts, cross-tenant denials, capacity sample, recovery and report QA.	QA: signed test log; developer: migration/rollback and updated guides.
24 Oct / 9.5	Owner and customer representative review defects and readiness; approve limited pilot only if gates pass.	Assim: go/no-go. If blockers remain, retain local/staging pilot and publish a corrective schedule.

Assumptions: dedicated developer availability plus QA support and 60-90 minutes of owner/domain review on key days. A few evening hours alone may not cover this plan. Do not trade missing acceptance evidence for a calendar commitment.

Full roadmap effort and resource assumptions

Provisional remaining effort: 55-75 developer days plus independent QA/domain review. At one dedicated developer and part-time QA, plan approximately 12-18 calendar weeks including contingency. Re-estimate after the first sprint.

Workstream mapped to original stages	Developer effort allowance	Dependency and accountable role
0-5 baseline and actual acceptance repairs	8-12 days	Current source and backup. Full-stack developer; QA; Assim and competent technical reviewers.
1/2/5 multi-industry templates and reports	6-8 days	Agreed workflow taxonomy and approved criteria. Developer + telecom/construction reviewers.
2 S06 bounded AI classification pilot	5-7 days	Labelled authorised dataset, provider decision and privacy/cost controls. Developer with AI experience.
6 Assets and handover	8-10 days	Accepted construction outputs, asset authority and receiver. Developer + asset owner.
7 Managed services	8-10 days	Accepted assets, real SLA and service process. Developer + operations reviewer.
8 Finance and closure	8-10 days	Contracts, certified quantities/service inputs and financial authority. Developer + finance reviewer.
9 hardening and controlled rollout	8-12 days	All released workflows, real-user availability, backups and operational owner. Developer + independent QA.
Contingency and integration reserve	4-6 days	Uncertain baseline defects, integrations, migration or infrastructure constraints. Assim controls scope.

Total allowance is a planning estimate, not a code-level delivery quotation. Workstreams can overlap selectively. Certification lead time, external audit fees, full offline sync, native mobile apps and full tower/CAD/digital-twin engines are outside these allowances.

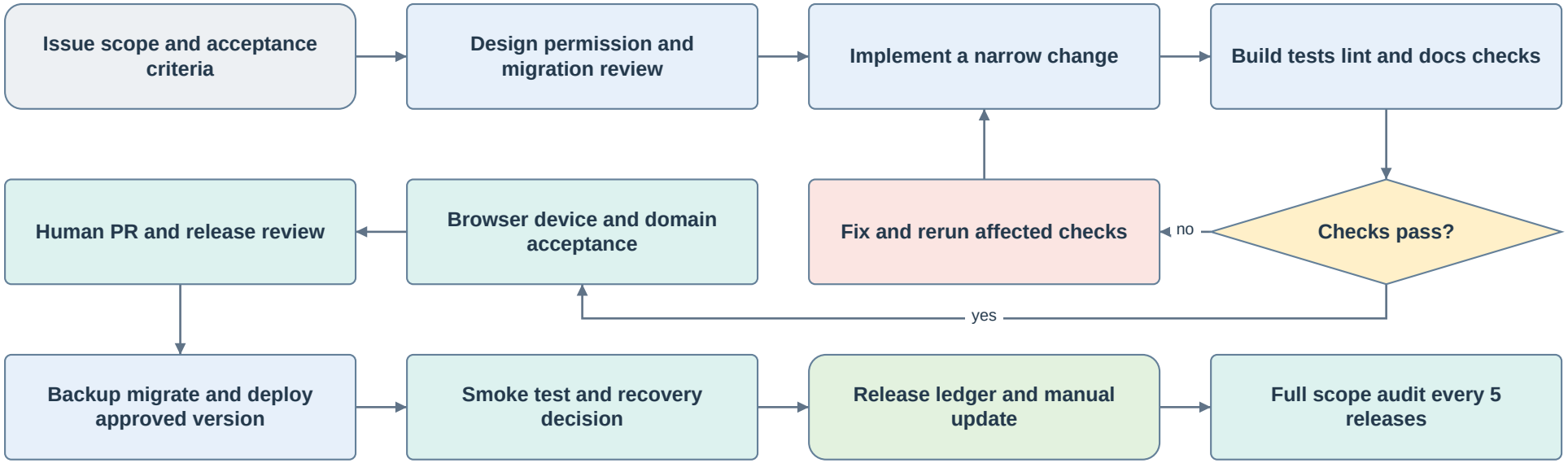
Implementation backlog and issue structure

Each issue must state original stage/substage, requirement IDs, input/output, permission impact, data change, tests and acceptance owner. Preserve existing issues121-128 and link new work rather than duplicating it.

Backlog priority	Stage and issue scope	Completion evidence
P0 Release truth	0 / 9.4: reconcile merged PR129, manual language and actual environment/deployment receipt.	Source SHA, environment identity and recorded publication/acceptance status agree.
P0 Evidence integrity	2.3 / 9.2: real upload limits, per-file retry and saved evidence linking.	No original loss or cross-Site association after interruption, retry and permission revocation.
P0 Workflow and permissions	2.4 / 5.2 / 9.1: independent review, return/retest and denied cross-role/tenant operations.	Three-person trace and customer authority; no blocked-work release.
P1 Navigation and templates	1 / 2 / 5: industry selector, inspection types, template publishing and practical reports.	Owner-approved screen flow and technically approved template per pilot workflow.
P1 Operational readiness	9.2-9.5: browser/mobile, load sample, restore, guides and pilot sign-off.	Evidence-based release decision; critical/high defects closed or release withheld.
P2 Advisory AI	2 S06: labelled dataset, bounded provider, group review and exception queue.	Measured task quality and time saved; manual path retained; no automatic pass.
P2 Lifecycle expansion	6 / 7 / 8: use queued issues125/126/127 with separate grants and staged acceptance.	Receiving, operational and financial decisions verified by the responsible people.

Release and change control workflow

Stage 9 / 9.1-9.5 implements operational acceptance. Human production approval and PR review follow the repository requirements.



Maintain manual, Site map, package status, project journey and release documentation. Run docs:generate and docs:check. Count actual successful functional releases after version44; every fifth release triggers the defined full-scope audit. Documentation-only work and unchanged redeployments do not count.

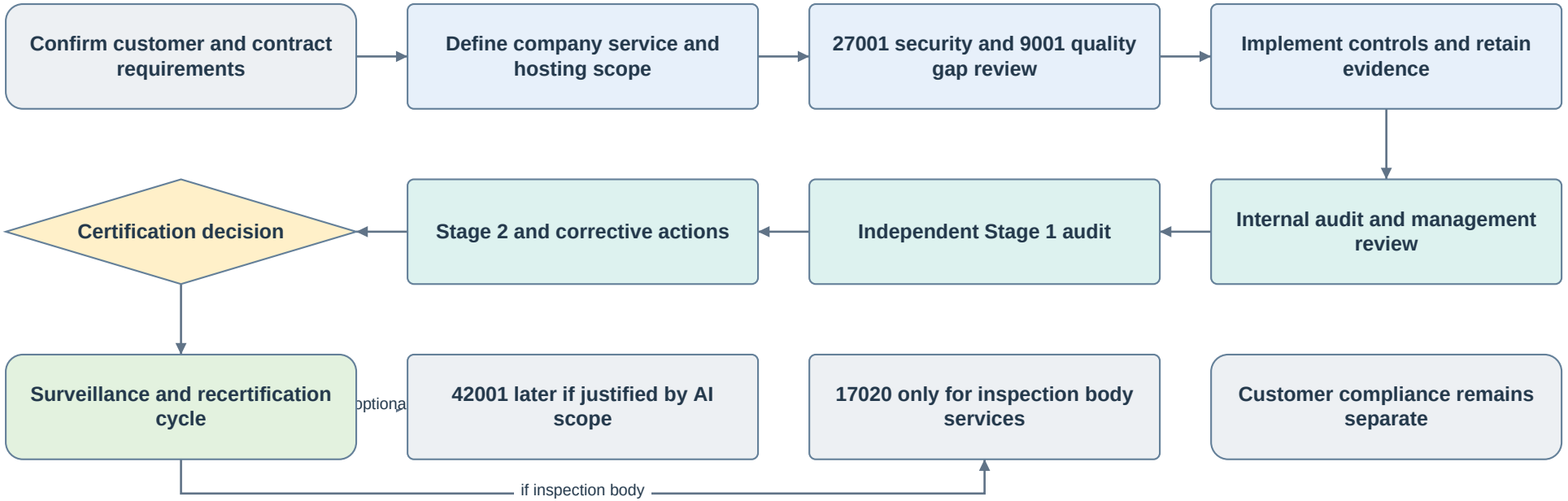
Stage 9 acceptance test matrix

Use reproducible scenarios and saved receipts. Historical source checks and synthetic fixtures remain separate from real field and customer acceptance.

Original substage	Mandatory scenarios	Gate owner
9.1 Lifecycle and authorisation	One telecom survey, one telecom inspection and one construction inspection; pass/return/reject/correct/stale-source cases. Role/layer/site and original-export denials.	QA + Assim; three independent workflow actors and customer representative where applicable.
9.2 Security and recovery	Bad files, parsing limits, direct-file scope, expired sessions, duplicate/racing writes, migration and restored DB plus original hashes.	Developer + independent security/QA reviewer. No accepted data loss or unauthorised access.
9.2 Capacity	200-photo representative real-device workflow; partial failures; agreed concurrent-user/dataset load; bounded AI job failure.	QA records p50/p95, file sizes/network and retry outcomes. Larger limits require a new accepted target.
9.3 Browser and device	Supported desktop browsers, Android/iPhone field use, keyboard/touch, focus, readable reports, navigation and errors.	Actual named users; viewport/device/version and observed evidence. HTTP checks alone are insufficient.
9.4 Documentation	Current scope, roles, limits, templates, instructions, routes, training and incident/recovery guide.	Document owner and release reviewer; generated docs checks and PDF visual review.
9.5 Controlled rollout	Pilot cohort, support contact, accepted scope, unresolved risks, rollback trigger and publication approval.	Assim plus responsible domain/customer owners. No automatic production release from CI success.

ISO certification and assurance workflow

ISO 27001 and ISO 9001 concern company management systems within defined scopes. Supporting a client checklist does not certify the client.



Inspection-body accreditation under ISO/IEC17020 is a separate business route. ISO does not issue company certificates. Verify the certificate issuer, accreditation scope and customer acceptance before paying; AI findings still require competent review.

ISO readiness requirements and budget position

Certification is a parallel business workstream, not a substitute for completing software acceptance.

Assurance route	Work products to prepare	Budget and decision
ISO/IEC 27001 priority	Defined app/hosting scope; risk assessment; Statement of Applicability; access, supplier, incident, development and recovery controls; internal audit/review records.	Published Pakistani small-company estimate: PKR650,000-1,200,000 for preparation and initial audit. Supplier indication only; taxes, technical fixes and recurring audits need a quote.
ISO 9001 next or integrated	Customer requirements, design/release controls, supplier review, competence, complaints, objectives, corrective actions and management review.	Published Pakistani small-company estimate: PKR120,000-200,000. Get a three-year itemised quotation and verify the applicable edition/transition with the issuer.
ISO/IEC 42001 later	AI inventory, accountability, impact/risk evaluation, dataset/provider controls, monitoring, human oversight and incidents.	No dependable Pakistan-specific quotation established here. Request pricing only when customer demand and AI scope justify it.
ISO/IEC 17020 conditional	Inspection-body competence, impartiality, methods, equipment/calibration, personnel and report authorisation.	PNAC accreditation route if the business itself performs inspections. Current fee and assessment scope must be quoted; software supply alone is different.

Cost references: Global Standards published ISO9001 and ISO27001 estimates checked18October2026; no provider accreditation endorsement is implied. Budget staff time, standards licences, training, security testing, surveillance and recertification separately. Do not use ISO logos or certification claims before an appropriate valid certificate exists.

Business launch and operating requirements

A usable product needs onboarding, support and measurable customer value alongside engineering features. Billing automation can follow initial validated pilots.

Requirement	Minimum release behavior	Measure or owner decision
Customer onboarding	Organisation/project setup, reviewed catalogs/templates, account assignments, sample training and source-data import checks.	Time to first completed reviewed inspection; support requests during first week.
Commercial model	Define organisation/project/user/storage entitlements and a bounded AI allowance. Keep professional inspection service charges separate.	Assim selects pilot terms and pricing after workload cost measurements; no invented market price.
Operating cost model	Storage GB-month + retrieval/egress + compute/jobs + AI usage + backups + support hours + assurance overhead.	Track cost per completed site/report and margin; 1,000 photos is a workload, not a pricing unit by itself.
Support and ownership	Named product owner, technical support path, incident severity, response window and ownership of content/records.	Published support promise matches staffed capability.
Usage and product metrics	Time from capture to approved report; first-pass acceptance; evidence gaps; reviewer effort; repeat use and cost/site.	Compare before/after on the pilot; avoid unvalidated time-saving or accuracy claims.
Pilot exit	Approved report, independent decisions, safe evidence retention, usability, support and customer willingness to continue.	Three workflow scenarios can be tested within one or more consenting organisations; do not invent pilot customers.

Risks dependencies and scope decisions

Resolve these items as part of Stage 0/9 planning. Routine reversible implementation can proceed within the agreed scope; external spending and production release remain explicit owner decisions.

Risk or dependency	Practical response	Owner and trigger
UI overload and unclear next step	Keep five primary navigation items; Site-specific workflow guidance; hide/disable unreleased tools honestly.	Assim + UX/developer; verify during Stage1/9.3 walkthrough.
Evidence loss or wrong Site	Immutable originals, digest checks, exact Site context, partial success and safe retries.	Developer + QA; any misassociation blocks release.
AI sorting creates false certainty	Use confirmed links, exception review, critical-miss metrics and manual fallback.	Technical reviewer; no automated engineering pass.
Scope expands beyond available time	Finish bounded pilot first. Implement originalStages6-8 only after foundation acceptance and resource commitment.	Assim; re-estimate when new workflow/integration is added.
Insufficient reviewer availability	Schedule L1/L2/L3 and customer participants before acceptance day; use distinct real identities.	Assim; unavailable reviewers keep acceptance pending.
Local and hosted drift	Record source SHA, migrations, configuration and deployment receipt; separate secrets from code.	Release owner; never infer production from merge.
Confidentiality and standard rights	Control external AI transfer, access, retention and licence use. Reference authorised criteria instead of copying entire standards.	Data owner + technical/commercial reviewer before provider enablement.

First development session runbook

Start with original Stage 0 / 0.1-0.4 and Stage 9 / 9.1 scope confirmation; then work through the dated sprint.

Sequence	Action	Concrete output
1 / Stage0	Identify the correct local repository and current branch; inspect pending changes and compare with main114ccf3e1273. Do not overwrite local work.	Source/environment baseline and list of local differences.
2 / Stage0	Create a dated external backup of source, databases, originals and required private configuration; perform a scoped restore check.	Backup manifest and recovery receipt; secrets stay private.
3 / Stage1 and9.3	Run the existing startup procedure; sign in with configured roles; walk assigned Sites and record navigation issues.	Annotated defect list and agreed three-workflow screen map.
4 / Stage2 and5	Configure one technically approved template per released workflow and nominate independent reviewers.	Template revisions, scope assignments and representative authorised data.
5 / Stage9	Create narrow issues with requirement IDs and tests; close P0 blockers before optional AI or future modules.	Ranked issue list mapped to existing stages and acceptance owners.
6 / Stage9.4-9.5	Update current status and guides after fixes; present the actual acceptance evidence and deployment package.	Reviewable release decision with scope, risks, rollback and publication status.

No repository changes, deployment, new integration, API credential setup, paid service or certification application was performed by this roadmap task. The editable diagrams are companion planning files; validate any later Visio conversion before relying on native editing fidelity.

Source register and planning limits

Evidence checked18October2026. Repository links are pinned to the inspected main commit where practical; access requires permission to the private repository.

Source	Reference
Project source baseline	https://github.com/assimbaig/isurvix-site-intelligence/commit/114ccf3e1273419b535be82f2724f7c0e6899064
Merged source delivery	https://github.com/assimbaig/isurvix-site-intelligence/pull/129
Successful latest inspected CI run	https://github.com/assimbaig/isurvix-site-intelligence/actions/runs/37979422864
Original stage register	https://github.com/assimbaig/isurvix-site-intelligence/blob/114ccf3e1273419b535be82f2724f7c0e6899064/docs/design/STAGE_STATUS.md
Current manual and package register	docs/SOFTWARE_MANUAL.md revision3.39; docs/PACKAGE_STATUS.md; docs/release-documentation.json at the baseline commit.
Repository controls and construction contracts	AGENTS.md; docs/SOFTWARE_REDESIGN_PLAN.md; docs/AUDIT_CHECKPOINT.json; packages/construction/src/policy.ts; package.json.
ISO and Pakistan accreditation	https://www.iso.org/certification.html | https://www.iso.org/standard/27001 | https://www.iso.org/standard/42001 | https://www.pnac.gov.pk/Services/Accreditation-Process
Published cost indications	https://globalstandards.com.pk/iso-9001-certification/ | https://globalstandards.com.pk/iso-27001-certification/

Evidence precedence: GitHub merge/CI facts update the stale draft-PR wording, but do not establish publication. Manual3.39 supersedes the older saved manual3.22 for planning. Source/read records support this roadmap; no fresh execution of the application or all test suites was performed. Proposed targets, estimates and new industry/template features are explicitly planning requirements.